

FORM U-1 MANUFACTURER'S DATA REPORT FOR PRESSURE VESSELS

Shop Order No.

National Board Number

As Required by the Provisions of the ASME Boiler and Pressure Vessel Code Rules, Section VIII, Division 1

1. Manufactured and certified by

(Name and address of Manufacturer)

2. Manufactured for

(Name and address of Purchaser)

3. Location of installation **NOT KNOWN**

(Name and address)

4. Type

Horizontal

(Horizontal, vertical, or sphere)

Heat Exchanger

(Tank, separator, jkt. vessel, heat exch., etc.)

(Manufacturer's serial number)

N/A

(CRN)

(Drawing number)

(National Board number)

2017

(Year built)

5. ASME Code, Section VIII, Div. 1

2015/ N/A

[Edition and Addenda, if applicable (date)]

N/A

(Code Case Number)

N/A

[Special Service per UG-120(d)]

Items 6-11 incl. to be completed for single wall vessels, jackets of jacketed vessels, shell of heat exchangers, or chamber of multichamber vessels

6. Shell: (a) Number of course(s) **1**

(b) Overall length

29' 5.0"

Course(s)			Material		Thickness		Long. Joint (Cat. A)			Circum. Joint (Cat. A, B, & C)			Heat Treatment	
No.	Diameter	Length	Spec./Grade or Type		Nom.	Corr.	Type	Full, Spot, None	Eff.	Type	Full, Spot, None	Eff.	Temp.	Time
1	4.3" OD	26' 3.0"	SA516-70		0.500	0.0625	1	None	70	1	None	70	1250F°	1.75HR
1	50.0" x 30.25"	3' 2.0"	SA516-70		0.500	0.0625	1	None	70	1	None	70	1250F°	1.75HR

Body Flanges on Shells

No.	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Location	Bolting			
									Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
1	WN	30.25"	36.5"	4.25	0.500	SA350 LF2	Butt	End	36 - 7/8"-11UNC	SA193-B7	N/A	N/A

7. Heads: (a)

SA516-70, 1250F, 1.75HR

(Material spec. number, grade or type) (H.T. - time and temp.)

(b)

N/A

(Material spec. number, grade or type) (H.T. - time and temp.)

	Location (Top, Bottom, Ends)	Thickness		Radius		Elliptical Ratio	Conical Apex Angle	Hemispherical Radius	Flat Diameter	Side to Pressure		Category A		
		Min.	Corr.	Crown	Knuckle					Convex	Concave	Type	Full, Spot, None	Eff.
(a)	End	0.1402	0.0625	N/A	N/A	2:1	N/A	N/A	N/A		X	N/A	N/A	N/A
(b)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A			N/A	N/A	N/A

Body Flanges on Heads

	Location	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Bolting			
									Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
(a)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		N/A	N/A	N/A

8. Type of jacket

N/A

Jacket closure

N/A

(Describe as ogee & weld, bar, etc.)

If bar, give dimensions; if bolted, describe or sketch

N/A

9. MAWP

50 psi

(Internal)

N/A

(External)

at max. temp.

300 °F

(Internal)

N/A

(External)

Min. design metal temp.

-20 °F

at

50 psi

10. Impact test

No, UG-20(f) Exemptat test temperature of **N/A**

(Indicate yes or no and the component(s) impact tested)

11. Hydro., pneu., or comb. test pressure

Hydro at 65 PSI

Proof test

N/A

Items 12 and 13 to be completed for tube sections.

12. Tubesheet

SA350 LF2

[Stationary (material spec. no.)]

36.5"

[Diameter (subject to press.)]

2.125

(Nominal thickness)

0.0625

(Corr. allow.)

Bolted

Attachment (welded or bolted)

N/A

[Floating (material spec. no.)]

N/A

(Diameter)

N/A

(Nominal thickness)

N/A

(Corr. allow.)

N/A

(Attachment)

13. Tubes

SA214

(Material spec. no., grade or type)

0.75

(O. D.)

14 BWG

(Nominal thickness)

352

(Number)

U

[Type (Straight or U)]

Items 14-18 incl. to be completed for inner chambers of jacketed vessels or channels of heat exchangers.

14. Shell: (a) No. of course(s) 1 (b) Overall length 1' 6.0"

Course(s)			Material		Thickness		Long. Joint (Cat. A)			Circum. Joint (Cat. A, B, & C)			Heat Treatment	
No.	Diameter	Length	Spec./Grade or Type		Nom.	Corr.	Type	Full, Spot, None	Eff.	Type	Full, Spot, None	Eff.	Temp.	Time
1	31.0" OD	1' 6.0"	SA516-70		0.375	0.0625	1	None	70	1	None	70	N/A	N/A

Body Flanges on Shells												
No	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Location	Bolting			
									Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
1	WN	30.25"	36.5"	4.625	0.375	SA350 LF2	Butt	End	36 - 7/8"-11UNC	SA193-B7	N/A	N/A

15. Heads: (a) SA516-70 (b) N/A
(Material spec. number, grade or type) (H.T. - time and temp.) (Material spec. number, grade or type) (H.T. - time and temp.)

	Location (Top, Bottom, Ends)	Thickness		Radius		Elliptical Ratio	Conical Apex Angle	Hemispherical Radius	Flat Diameter	Side to Pressure		Category A		
		Min.	Corr.	Crown	Knuckle					Convex	Concave	Type	Full, Spot, None	Eff.
(a)	END	0.1985	0.0625	N/A	N/A	2:1	N/A	N/A	N/A		X	N/A	N/A	N/A

Body Flanges on Heads												
	Location	Type	ID	OD	Flange Thk	Min Hub Thk	Material	How Attached	Bolting			
									Num & Size	Bolting Material	Washer (OD, ID, thk)	Washer Material
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

16. MAWP 150 psi N/A at max. temp. 500 °F N/A Min. design metal temp. -20 °F at 150 psi
(Internal) (External) (Internal) (External)17. Impact test No, UG-20(f) Exempt at test temperature of N/A
(Indicate yes or no and the component(s) impact tested)18. Hydro., pneu., or comb. test pressure Hydro. at 195 PSI Proof test N/A

19. Nozzles, inspection, and safety valve openings:

Purpose (Inlet, Outlet, Drain, etc.)	No.	Diameter or Size	Type	Material		Nozzle Thickness		Reinforcement Material	Attachment Details		Location (Insp. Open.)
				Nozzle	Flange	Nom.	Corr.		Nozzle	Flange	
Inlet/Outlet	3	10	300# RFWN	SA106B	SA105	0.365	0.0625	None	(c)	Butt	
Outlet	1	20	150# RFWN	SA106B	SA105	0.375	0.0625	None	(c)	Butt	
Outlet	1	8	150# RFWN	SA106B	SA105	0.322	0.0625	None	(c)	Butt	
Bridle/Drain	6	2	150# RFWN	SA106B	SA105	0.218	0.0625	None	(c)	Butt	
TE/TI/PI	3	3/4	3000# HC	SA105		NPT	0.0625	None	(bb)		

20. Supports: Skirt No Lugs N/A Legs Two Others N/A Attached Shell, Welded
(Yes or no) (Number) (Number) (Describe) (Where and how)

21. Manufacturer's Partial Data Reports properly identified and signed by Commissioned Inspectors have been furnished for the following items of the report (list the name of part, item number, Manufacturer's name, and identifying number):

N/A

22. Remarks

Item #19: Nozzle Attachment per Fig. UW-16.1For non-lethal service.Over-pressure protection provided by others.

CERTIFICATE OF SHOP COMPLIANCE

We certify that the statements in this report are correct and that all details of design, material, construction, and workmanship of this vessel conform to the ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. U Certificate of Authorization Number [REDACTED] Expires [REDACTED]

Date [REDACTED] Name [REDACTED] Signed [REDACTED]
(Manufacturer) (Representative)

CERTIFICATE OF SHOP INSPECTION

I, the undersigned, holding a valid commission issued by the National Board of Boiler and Pressure Vessel Inspectors and employed by [REDACTED]

have inspected the pressure vessel described in this Manufacturer's Data Report on [REDACTED], and state that,

to the best of my knowledge and belief, the Manufacturer has constructed this pressure vessel in accordance with ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. By signing this certificate neither the Inspector nor his/her employer makes any warranty, expressed or implied, concerning the pressure vessel described in this Manufacturer's Data Report. Furthermore, neither the Inspector nor his/her employer shall be liable in any manner for any personal injury or property damage or a loss of any kind arising from or connected with this inspection.

Date [REDACTED] Signed [REDACTED] Commissions: [REDACTED]
(Authorized Inspector) [National Board (incl. endorsements)]

CERTIFICATE OF FIELD ASSEMBLY COMPLIANCE

We certify that the statements made in this report are correct and that the field assembly construction of all parts of this vessel conforms with the requirements of ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. U Certificate of Authorization Number [REDACTED] Expires [REDACTED]

Date [REDACTED] Name [REDACTED] Signed [REDACTED]
(Assembler) (Representative)

CERTIFICATE OF FIELD ASSEMBLY INSPECTION

I, the undersigned, holding a valid commission issued by The National Board of Boiler and Pressure Vessel Inspectors and employed by [REDACTED]

have compared the statements in this Manufacturer's Data Report with the described pressure vessel and state that parts referred to as data items [REDACTED], not included in the certificate of shop inspection, have been inspected by me and to the best of my knowledge and belief, the Manufacturer has constructed and assembled this pressure vessel in accordance with the ASME BOILER AND PRESSURE VESSEL CODE, Section VIII, Division 1. The described vessel was inspected and subjected to a hydrostatic test of [REDACTED]. By signing this certificate neither the Inspector nor his/her employer makes any warranty, expressed or implied, concerning the pressure vessel described in this Manufacturer's Data Report. Furthermore, neither the Inspector nor his/her employer shall be liable in any manner for any personal injury or property damage or a loss of any kind arising from or connected with this inspection.

Date [REDACTED] Signed [REDACTED] Commission [REDACTED]
(Authorized Inspector) [National Board (incl. endorsements)]